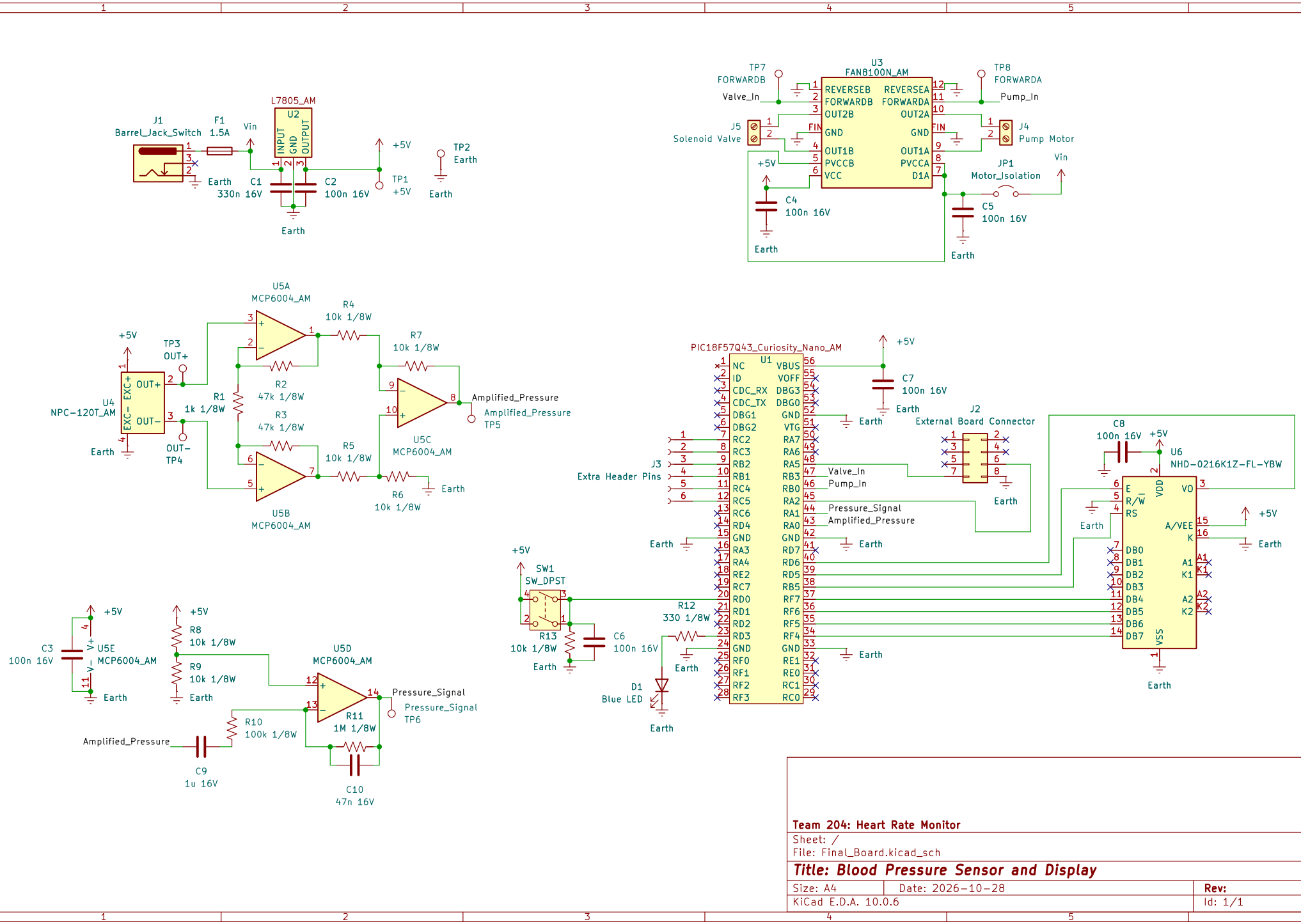

J1
Barrel_Jack_Switch
F1
1.5A
Vin
L7805_AM
U2
INPUT
GND
OUTPUT
+5V
TP2
Earth
TP1
+5V
Earth
C1
330n 16V
C2
100n 16V
Earth
TP7
FORWARDB
Valve_In
U3
FAN8100N_AM
REVERSEB
FORWARDB
OUT2B
GND
OUT1B
PVCCB
VCC
REVERSEA
FORWARDA
OUT2A
GND
OUT1A
PVCCA
D1A
TP8
FORWARDA
Pump_In
J5
Solenoid Valve
J4
Pump Motor
JP1
Motor_Isolation
Vin
+5V
C4
100n 16V
Earth
C5
100n 16V
Earth
U5A
MCP6004_AM
R4
10k 1/8W
R7
10k 1/8W
+5V
TP3
OUT+
U4
NPC-120T_AM
EXC+
EXC-
OUT+
OUT-
Earth
OUT-
TP4
R1
1k 1/8W
R2
47k 1/8W
R3
47k 1/8W
R5
10k 1/8W
R6
10k 1/8W
Earth
U5B
MCP6004_AM
U5C
MCP6004_AM
Amplified_Pressure
Amplified_Pressure
TP5
PIC18F57Q43_Curiosity_Nano_AM
U1
NC
ID
CDC_RX
CDC_TX
DBG1
DBG2
RC2
RC3
RB2
RB1
RC4
RC5
RC6
RD4
GND
RA3
RA4
RE2
RC7
RD0
RD1
RD2
RD3
GND
RF0
RF1
RF2
RF3
VBUS
VOFF
DBG3
DBG0
GND
VTG
RA7
RA6
RA5
RB3
RB0
RA2
RA1
RA0
GND
RD7
RD6
RD5
RB5
RF7
RF6
RF5
RF4
GND
RE1
RE0
RC1
RC0
+5V
C7
100n 16V
Earth
J3
Extra Header Pins
Valve_In
Pump_In
Pressure_Signal
Amplified_Pressure
Earth
J2
External Board Connector
Earth
C8
100n 16V
+5V
U6
NHD-0216K1Z-FL-YBW
E
R/W
RS
VDD
VO
A/VEE
K
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
VSS
A1
K1
A2
K2
+5V
Earth
Earth
SW1
SW_DPST
R12
330 1/8W
R13
10k 1/8W
C6
100n 16V
Earth
D1
Blue LED
Earth
C3
100n 16V
+5V
U5E
MCP6004_AM
Earth
R8
10k 1/8W
R9
10k 1/8W
Earth
U5D
MCP6004_AM
Pressure_Signal
Pressure_Signal
TP6
R10
100k 1/8W
R11
1M 1/8W
Amplified_Pressure
C9
1u 16V
C10
47n 16V
Team 204: Heart Rate Monitor
Sheet: /
File: Final_Board.kicad_sch
Title: Blood Pressure Sensor and Display
Size: A4
Date: 2026-10-28
Rev:
KiCad E.D.A. 10.0.6
Id: 1/1